

Ram Binay Gupta

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SUMMARY

Research oriented M.Sc. Computer Science student at Philipps Universität Marburg with a strong background in algorithms, distributed systems, and machine learning. Experienced in scalable data processing, Software Engineering, Retrieval Augmented Generation (RAG), and AI driven knowledge systems. Skilled in designing and developing intelligent and data intensive systems for interdisciplinary research applications, with strong analytical and programming expertise and interests in optimization, and designing scalable, and human centered AI systems.

Primary Stack: Python, Java, SQL, Spark, PyTorch, RAG Systems, LLM based Systems.

EDUCATION

04/2024 – Present	M.Sc. Computer Science , Philipps Universität Marburg, Germany Major: ML, XAI, NLP, Big Data System, Geo Database
08/2014 – 06/2016	Master Computer Application , Lovely Professional University, India Cumulative Grade: 9.52 / 10.0 Major: Software Programming, Computer Based Optimization, Data Mining
08/2009 – 01/2014	Bachelor Computer Application , Pokhara University, Nepal Cumulative Grade: 3.98 / 4.0

PROFESSIONAL EXPERIENCE

10/2025 – Present	Student Assistant , Multimodal Modelling & Learning Group / Databases Group / Big Data Analytics Group, Philipps Universität Marburg - Contributed to the CuRAG research project involving AI assisted knowledge graph curation with human-in-the-loop validation. - Conducted tutorial sessions and evaluated assignments and projects for the Data Integration, NoSQL courses.
06/2025 – 07/2025	Tutor , Philipps Universität Marburg Designed and delivered tutorial sessions on Python, LaTeX, Git, and AI tools for research to international students at the FB12 department.
04/2018 – 10/2023	Assistant Lecturer , Chitwan College of Technology, Nepal - Designed plans and took computer science courses for bachelor students. - Supervised project work and graded practical task and examination.
05/2018 – 04/2020	Assistant Software Developer , Crystal Light IT Pvt. Ltd., Nepal - Developed and tested Java based software applications, contributing to implementation, and debugging activities. - Provided technical support to end users to resolve user problems.

PROJECTS

CuRAG	• Designing and developing interfaces enabling domain experts to curate and validate a Neo4j based knowledge graph, while also supporting question answering grounded in attached PDF documents using local LLMs. • Implementing workflows for modifying nodes, relationships, and metadata. • Collaborating in the team to deliver innovative solution in a research driven project environment. • Skills & Tools: Full Stack Web Development using React, Python, Graph Database using Neo4j.
Historical RAG System	Speech • Developed a RAG pipeline to retrieve relevant historical speech transcripts and generate context aware responses. • Processed and cleaned transcript datasets, structured them for efficient retrieval and built embedding vector stores. • Integrated retrieval results with an LLM. • Designed evaluation tests to measure retrieval precision and semantic relevance. • Skills & Tools: Python, Elastic Search, Information retrieval, Vector databases.
Image Classification Using CNN	• Developed image classification models using PyTorch and CNN architectures. • Trained and evaluated models on labeled image datasets using standard deep learning workflows and achieved accuracy of 0.94. • Applied pre-processing, augmentation, and performance evaluation. • Skills & Tools: Python, PyTorch, Deep Learning, CNNs, Data Pre-processing.

TECHNICAL SKILLS

Prog. Languages	Python, Java, C, C#
Distributed Systems	Apache Spark, Akka, Apache Sedona
Databases	SQL, PostgreSQL, Neo4j, Redis, MongoDB
AI & ML Systems	PyTorch, RAG Systems, LLM Agents, LangChain, Transformers, NLP
Tools	Git, Github, Docker, LaTeX

ACHIEVEMENTS

05/2017	University Academic Honour , Lovely Professional University Received the University Academic Honour award for standing First in order of merit in Master of Computer Applications Program in the class of 2016.
04/2014	Dean's List Award , Pokhara University Received the Dean's List Award in recognition of an outstanding meritorious achievement in Bachelor of Computer Application.

